

CHE 201 Spring 2020 HW 2

Thanh Nguyen

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 20 / 20

✓ - **0 pts** Correct

- **5 pts** Effect of friction is wrong or not answered.
- **5 pts** Calculation of constant force R is wrong.
- **5 pts** Calculation of the force of gravity is wrong.
- **5 pts** Energy balance is wrong.

QUESTION 2

2 Question 2 20 / 20

- **0 pts** Correct
- **5 pts** Wrong calculation in b)
- ✓ - **0 pts** Be more precise on calculation. Keep more significant digits.
- **5 pts** Wrong equation in b)
- **5 pts** Wrong calculation in a)
- **5 pts** Wrong equation in a)

QUESTION 3

3 Question 3 20 / 20

- ✓ - **0 pts** Correct
- **4 pts** Wrong sketch of the processes.
- **4 pts** Wrong answer in a)
- **4 pts** Wrong 1-2 work in b)
- **4 pts** Wrong 2-3 work in b)
- **0 pts** Please distinguish a positive or a negative sign to denote the energy transfer direction by work.
- **4 pts** Wrong 3-1 work in b)

QUESTION 4

4 Question 4 20 / 20

- ✓ - **0 pts** Correct
- **5 pts** Wrong equation used in getting power
- **5 pts** Wrong calculation in getting power.
- **5 pts** Wrong equation used in getting the speed

- **5 pts** Wrong calculation in getting the speed

QUESTION 5

5 Question 5 20 / 20

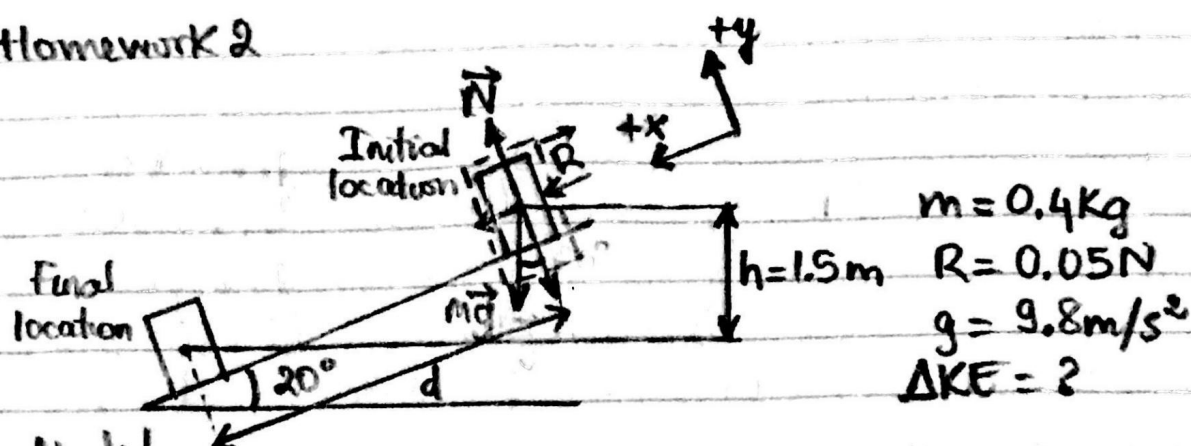
- ✓ - **0 pts** Correct
- **2.5 pts** Not enough current federal regulations.
- **2.5 pts** Not enough current regulations in state and local area.
- **5 pts** Not enough comments on the effectiveness of services.
- **20 pts** Question 5 not answered.
- **5 pts** No current federal regulations.
- **2 pts** No reference added.
- **5 pts** No current regulations in state or local area.

Thanh Nguyen

CHE 201 - Homework 2

1.)

Sketch



Engineering Model

1. The can is a closed system
2. $g = 9.8 \text{ m/s}^2$
3. There is no friction between the can and the surface
4. Define W on the system > 0
5. Define the $+x$ axis and $+y$ axis as described on the sketch

Analysis

a. In the x direction: $\Sigma F_x = mg \sin 20^\circ + R$

$$\rightarrow W \text{ done by } \Sigma F_x: W_x = (\Sigma F_x) d = (\Sigma F_x) \frac{h}{\sin 20^\circ}$$

Since there is no work done by ΣF_y , W_{total} on the system is W_x

$$\text{Therefore, } \Delta KE = W_{\text{total}} = W_x \\ = (mg \sin 20^\circ + R) \left(\frac{h}{\sin 20^\circ} \right)$$

$$= [(0.4 \text{ kg})(9.8 \text{ m/s}^2) \sin 20^\circ + 0.05 \text{ N}] \left(\frac{1.5 \text{ m}}{\sin 20^\circ} \right) = 6.10 \text{ J}$$

$\rightarrow$ Kinetic energy is increasing as the can moves from the initial to final location.

b. If friction between the can and the inclined surface were significant, the value of the change in kinetic energy would be less than the value calculated in part a.

1 Question 1 20 / 20

✓ - 0 pts Correct

- 5 pts Effect of friction is wrong or not answered.
- 5 pts Calculation of constant force R is wrong.
- 5 pts Calculation of the force of gravity is wrong.
- 5 pts Energy balance is wrong.

2.)

Sketch



$$\begin{aligned} P_i &= 1 \text{ bar} \\ V_i &= 0.1 \text{ m}^3 \\ P_f &= 9 \text{ bar} \\ V_f &= ? ; W = ? \end{aligned}$$

Engineering Model

1. The gas is a closed system
2. $pV^2 = \text{constant}$

Analysis

$$a. P_i V_i^2 = P_f V_f^2 \rightarrow V_f^2 = \frac{P_i V_i^2}{P_f} \rightarrow V_f = \sqrt{\frac{P_i V_i^2}{P_f}} = \sqrt{\frac{(1 \text{ bar})(0.1 \text{ m}^3)^2}{9 \text{ bar}}} = 0.03 \text{ m}^3$$

$$b. \text{Constant} = P_i V_i^2 = (1 \text{ bar})(0.1 \text{ m}^3)^2 = 0.01 \text{ bar} \cdot \text{m}^6$$

$$W = \int_{V=0.1}^{V=0.03} p dV = \int_{V=0.1}^{V=0.03} \frac{0.01}{V^2} dV = \frac{-0.01}{V} \bigg|_{V=0.1}^{V=0.03} = \frac{-0.01}{0.03} + \frac{0.01}{0.1} = -0.23 \text{ bar} \cdot \text{m}^3$$

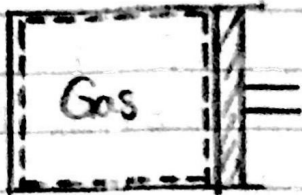
$$W = -0.23 \text{ bar} \cdot \text{m}^3 \times \frac{10^5 \text{ N/m}^2}{1 \text{ bar}} = -23000 \text{ N} \cdot \text{m} = -23000 \text{ J} = -23 \text{ kJ}$$

2 Question 2 20 / 20

- 0 pts Correct
- 5 pts Wrong calculation in b)
- ✓ - 0 pts Be more precise on calculation. Keep more significant digits.
- 5 pts Wrong equation in b)
- 5 pts Wrong calculation in a)
- 5 pts Wrong equation in a)

3.)

Sketch



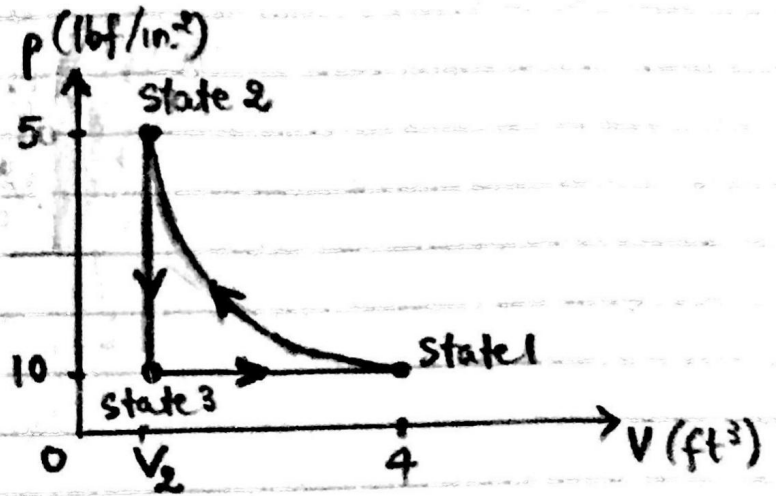
$$P_1 = 10 \text{ lbf/in}^2; V_1 = 4 \text{ ft}^3$$

$$P_2 = 50 \text{ lbf/in}^2$$

Process 1-2: $pV = \text{constant}$

Process 2-3: constant volume

Process 3-1: constant pressure



Engineering Model

1. The gas is a closed system

2. Process 1-2: $pV = \text{constant}$

Analyses

$$a. P_1 V_1 = P_2 V_2 \rightarrow V_2 = \frac{P_1 V_1}{P_2} = \frac{(10 \text{ lbf/in}^2)(4 \text{ ft}^3)}{50 \text{ lbf/in}^2} = 0.8 \text{ ft}^3$$

$$b. W_1 = \int_{V=4}^{V=0.8} p dV = \int_{V=4}^{V=0.8} \frac{\text{constant}}{V} dV = \int_{V=4}^{V=0.8} \frac{P_1 V_1}{V} dV = \int_{V=4}^{V=0.8} \frac{40}{V} dV = 40 \ln V \Big|_{V=4}^{V=0.8}$$

$$= 40 (\ln 0.8 - \ln 4) = -64.4 (\text{lbf/in}^2)(\text{ft}^3)$$

$$W_1 = -64.4 (\text{lbf} \cdot \text{ft}^3 / \text{in}^2) \times \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 \times \frac{0.00129 \text{ Btu}}{1 \text{ lbf} \cdot \text{ft}} = -11.96 \text{ Btu}$$

$$W_2 = 0 \text{ Btu}$$

$$W_3 = p \Delta V = (10 \text{ lbf/in}^2)(4 \text{ ft}^3 - 0.8 \text{ ft}^3) = 32 (\text{lbf/in}^2)(\text{ft}^3)$$

$$W_3 = 32 (\text{lbf} \cdot \text{ft}^3 / \text{in}^2) \times \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 \times \frac{0.00129 \text{ Btu}}{1 \text{ lbf} \cdot \text{ft}} = 5.94 \text{ Btu}$$

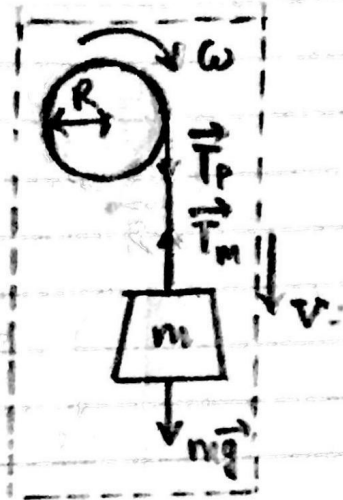
3 Question 3 20 / 20

✓ - 0 pts Correct

- 4 pts Wrong sketch of the processes.
- 4 pts Wrong answer in a)
- 4 pts Wrong 1-2 work in b)
- 4 pts Wrong 2-3 work in b)
- 0 pts Please distinguish a positive or a negative sign to denote the energy transfer direction by work.
- 4 pts Wrong 3-1 work in b)

4.)

Sketch



$$\begin{aligned} R &= 3 \text{ in.} \\ v &= 5 \text{ ft/s} \\ g &= 32.2 \text{ ft/s}^2 \\ m &= 5 \text{ lb.} \\ P &= ? ; \omega = ? \end{aligned}$$

Engineering Model

1. It is a closed system
2. $g = 32.2 \text{ ft/s}^2$
3. The rope does not slip, and it is massless.

Analysis

a. Since the block moves at a constant speed, there is no net force applied to the block. This means: $mg = T_m$

Also, since the rope is massless, $T_m = T_p$

Power transmitted to the pulley is

$$\begin{aligned} P &= T_p \cdot v = mgv = (5 \text{ lb}) (32.2 \text{ ft/s}^2) (5 \text{ ft/s}) = 805 \text{ lb} \cdot \text{ft}^2/\text{s}^3 \\ P &= 805 \text{ lb} \cdot \text{ft}^2/\text{s}^3 \times \frac{0.45 \text{ kg}}{1 \text{ lb}} \times \left(\frac{0.305 \text{ m}}{1 \text{ ft}} \right)^2 = 33.7 \text{ kgm}^2/\text{s}^3 = 33.7 \text{ W} \end{aligned}$$

$$P = 33.7 \text{ W} \times \frac{1 \text{ hp}}{745.7 \text{ W}} = 0.045 \text{ hp}$$

$$b. \omega = \frac{v}{R} = \frac{5 \text{ ft/s}}{3 \text{ in}} = \frac{5 \text{ ft/s}}{3 \text{ in}} \times \frac{12 \text{ in}}{1 \text{ ft}} = 20 \text{ rad/s}$$

$$\omega = 20 \text{ rad/s} \times \frac{1 \text{ rev.}}{2\pi \text{ rad}} \times \frac{60 \text{ s}}{1 \text{ min}} = 191 \text{ RPM}$$

4 Question 4 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong equation used in getting power
- 5 pts Wrong calculation in getting power.
- 5 pts Wrong equation used in getting the speed
- 5 pts Wrong calculation in getting the speed

Federal, State, and Local Regulations on Battery Collection and Management

Federal Regulations on Used Battery Collection and Management

Battery collection and management is regulated by the United States Environmental Protection Agency (EPA), the 1996 Battery Act, and the Universal Waste Rule. There are two main sections in the Battery Act. The first section, title I, facilitates the recycling of Nickel Cadmium, Small Sealed Lead Acid (SSLA), lead-acid and other rechargeable batteries, which are all referred to as “regulated batteries” (EPA, pg. 7). All regulated batteries must be labeled the name of the material that batteries contain—for example, “Pb” for lead-acid batteries, “NiCd” for Nickel Cadmium batteries—and “BATTERY MUST BE RECYCLED OR DISPOSED OF PROPERLY” (pg. 7-8). The second section, title II, phases out the use of batteries that contain mercury. This title prohibits the sales of alkaline-manganese and zinc-carbon batteries that contain mercury and mercuric-oxide batteries, except for some certain reasons (pg. 9). Using the Universal Waste Rule, the Battery Act implies, regarding used battery management, that from title 40, part 273.13 and part 273.33 (eCFR):

1. A person who accumulates any amount of universal waste batteries is prohibited from diluting and disposing of universal waste. Regardless of the amount of universal waste batteries, a handler of universal waste must contain any universal waste battery that shows evidence of leakage, spillage, or damage that could cause leakage under reasonably foreseeable conditions in a container. The container must be closed, structurally sound, compatible with the contents of the battery, and must lack evidence of leakage, spillage, or damage that could cause leakage under reasonably foreseeable conditions.
2. If the casing of each individual battery cell is not breached and remains intact and closed, the handler can sort batteries by type, mix battery types in one container, discharge batteries so as to remove the electric charge, regenerate used batteries, disassemble batteries or battery packs into individual batteries or cells, remove batteries from consumer products, or remove electrolyte from batteries.
3. A handler who removes electrolyte from batteries or generates other solid waste as a result of the activities listed above must determine whether the electrolyte and/or other solid waste exhibit a characteristic of hazardous waste in order to have specific solutions to clean up the waste.

Illinois Regulations on Used Battery Collection and Management

The state of Illinois currently does not have any specific battery regulations. Therefore, the federal regulations for battery management apply in Illinois (Call2Recycle).

Local Regulations on Used Battery Collection and Management: Northern Cook County

Battery management in the Northern Cook County is regulated by the Solid Waste Agency of Northern Cook County (SWANCC). According to SWANCC, single-use, non-rechargeable alkaline batteries AAA, AA, C, D 6V and 9V are classified as “non-hazardous” and may be safely disposed of with normal household waste. Rechargeable batteries—including Nickel Cadmium, Nickel Metal Hydride, Lithium, Zinc-Air, Silver Oxide, and SSLA—contain mercury and other heavy metals and should be recycled (SWANCC). Lead-acid batteries in automotives, marines, and sump pumps are prohibited from being disposed of in the trash (SWANCC). These batteries need to be brought to special outlets for recycling (SWANCC).

Comments on the Effectiveness of Services Working to Assist Consumers in Complying with those Regulations

Although there are specific regulations regarding battery collection and management, it is still not very effective because batteries have not been commonly recycled, and the citizens are not familiar with all different kinds of batteries. In addition, there are not many public-education programs nor committees about waste sorting and recycling, especially batteries. There are very few particular promotions of rechargeable battery recycling in the local area and the state. In order to improve the effectiveness of services, the state and local governments should create regular committees for people to come and learn more about different types of batteries and different ways to collect and manage used batteries. Besides, schools should also give students at least a quick look at how important used battery management is to the environment and teach them how to “dispose” battery waste properly.

References

1. “Implementation of the Mercury-Containing and Rechargeable Battery Management Act.” United States Environmental Protection Agency (EPA), Nov. 1997, URL: www.call2recycle.org/wp-content/uploads/ImplementationoftheMercury-ContainingandRechargeableBatteryManagementAct.pdf. Accessed 1 Feb. 2020.
2. “Batteries.” Solid Waste Agency of Northern Cook County (SWANCC), 12 Sep. 2018, URL: www.swancc.org/batteries. Accessed 2 Feb. 2020.
3. “Recycling Laws by State.” Call2Recycle, URL: www.call2recycle.org/recycling-laws-by-state/#Illinois. Accessed 2 Feb. 2020.
4. “Title 40: Protection of Environment - Part 273.” Electronic Code of Federal Regulations (eCFR), 30 Jan. 2020, URL: ecfr.gov/cgi-bin/text-idx?SID=0501d91ec562faafa833c60c2404d806&mc=true&node=pt40.27.273&rgn=div5#sp40.29.273. Accessed 2 Feb. 2020.

5 Question 5 20 / 20

✓ - 0 pts Correct

- 2.5 pts Not enough current federal regulations.
- 2.5 pts Not enough current regulations in state and local area.
- 5 pts Not enough comments on the effectiveness of services.
- 20 pts Question 5 not answered.
- 5 pts No current federal regulations.
- 2 pts No reference added.
- 5 pts No current regulations in state or local area.